seventy years. Its light output had changed substantially within periods as short as a month. This means that a large portion of the light from 3C273 must come from a region smaller than the distance light travels in a month, that is, smaller than 1 “light-month.” (If the region were larger, then there would be no way that any force, traveling, of course, at a speed less than or equal to that of light, could make the emitting gas all brighten up or dim out simultaneously to within an accuracy of a month.)

The implications were extremely hard to believe. This weird quasar, this 3C273, was shining 100 times more brightly than the brightest galaxies in the Universe; but whereas galaxies produce their light in regions 100,000 light-years in size, 3C273 produces its light in a region at least a million times smaller in diameter and 10^{18} times smaller in volume: just a light-month or less. The light must come from a massive, compact, gaseous object that is heated by an enormously powerful engine. The engine would ultimately turn out to be, with high but not complete confidence, a gigantic black hole, but strong evidence for this was still fifteen years into the future.

If explaining Jansky’s radio waves from our own Milky Way galaxy was difficult, and explaining the radio waves from distant radio galaxies was even more difficult, then the explanation for radio waves from these superdistant quasars would have to be superdifficult.

The difficulty, it turned out, was an extreme mental block. Jesse Greenstein, Fred Whipple, and all other astronomers of the 1930s and 1940s had presumed that cosmic radio waves, like light from stars, are emitted by the heat-induced jiggling of atoms, molecules, and electrons. Astronomers of the thirties and forties could not conceive of any other way for nature to create the observed radio waves, even though their calculations showed unequivocally that this way can’t work.

Another way, however, had been known to physicists since the early twentieth century: When an electron, traveling at high speed, encounters a magnetic field, the field’s magnetic force twists the electron’s motion into a spiral. The electron is forced to spiral around and around the magnetic field lines (Figure 9.4), and as it spirals, it emits electromagnetic radiation. Physicists in the 1940s began to call this radiation *synchrotron radiation,* because it is produced by spiraling electrons in the particle accelerators called “synchrotrons” that they were then building. Remarkably, in the